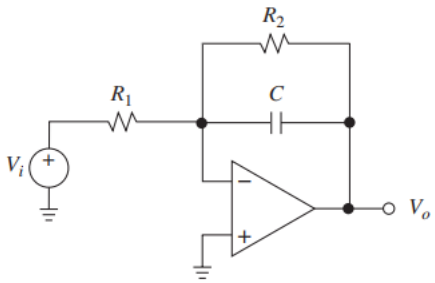
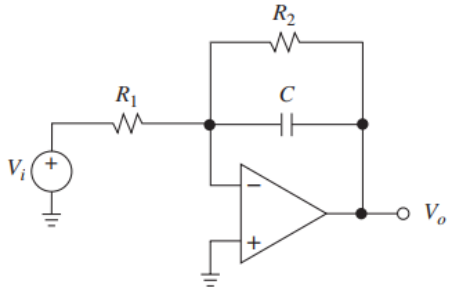
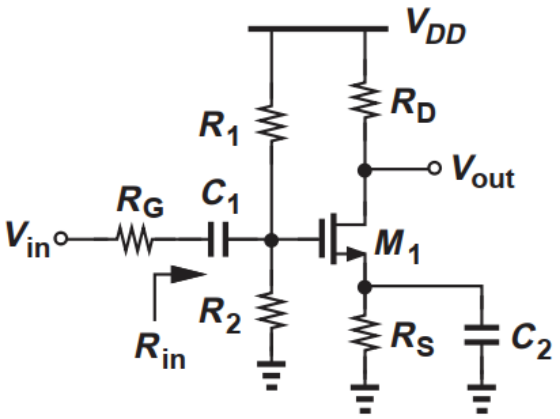
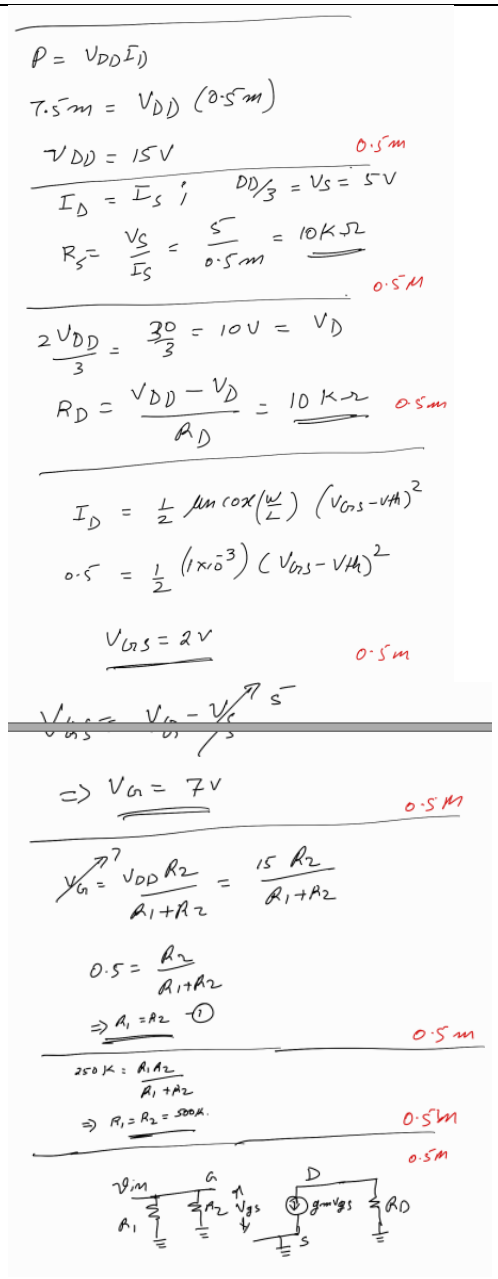
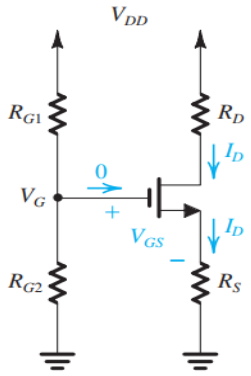
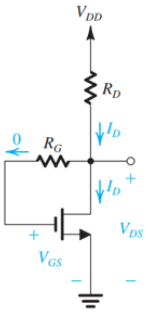
ECE 2126 ANALOG CIRCUITS MID TERM SCHEME JULY-DEC 2025
TOT MARKS:30

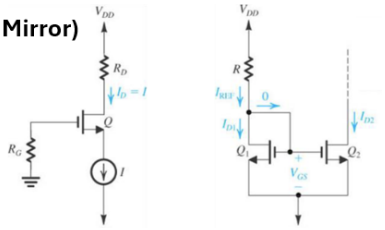
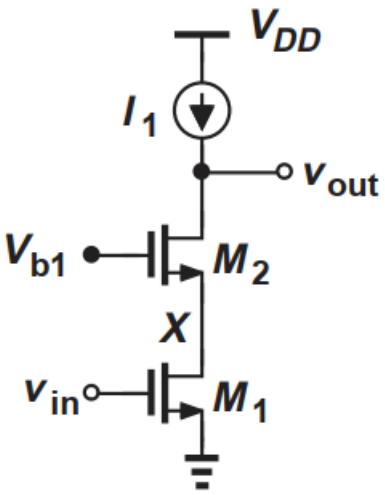
Q. No	Question	Mark	CLO
Q1	An ideal op-amp has an infinite open-loop gain ($a \to \infty$). A consequence of this is that the differential input voltage, V_D, approaches zero for any finite output voltage V_0. How can a non-zero output voltage be sustained if the input voltage difference is zero 1. The common-mode input voltage becomes infinitely large to compensate. 2. The output voltage is sustained by the op-amp's power supplies, independent of the input. 3. **As 'a' approaches infinity, vD approaches zero in such a way that the product 'a * V_D ' remains finite and equal to V_0. 4. The input bias currents provide a small offset that is magnified to the final output voltage.	**0.5**	**4**
Q2	For the circuit shown in the figure, at what frequency the 3-dB frequency, occur?  1. When the magnitude of the capacitive impedance equals the input resistance (|Zc| = R_1). 2. When the open-loop gain of the op-amp begins to roll off. 3. **When the magnitude of the capacitive impedance equals the feedback resistance (|Zc| = R_2). 4. When the input and feedback resistances are equal ($R_1 = R_2$).	**0.5**	**4**
	Design a summing amplifier that produces the output V0 = -(4V1 + 2.5V2 + 10V3). Using standard 5% resistor values, a possible component set is 1. R_1=10kΩ, R_2=16kΩ, R_3=4kΩ, RF=40kΩ 2. R_1=12kΩ, R_2=20kΩ, R_3=5.1kΩ, RF=51kΩ 3. **R_1=30kΩ, R_2=47kΩ, R_3=12kΩ, RF=120kΩ 4. R_1=10kΩ, R_2=20kΩ, R_3=5kΩ, RF=40kΩ	**0.5**	**4**

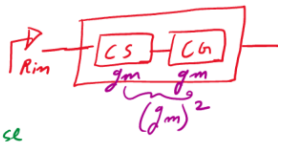
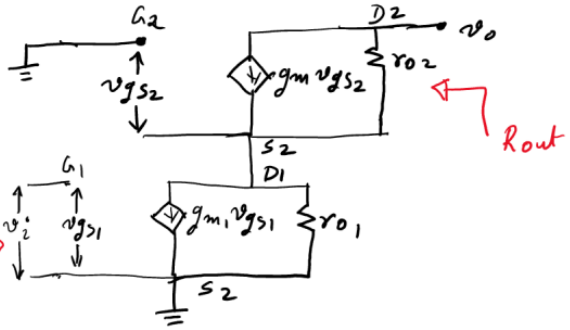
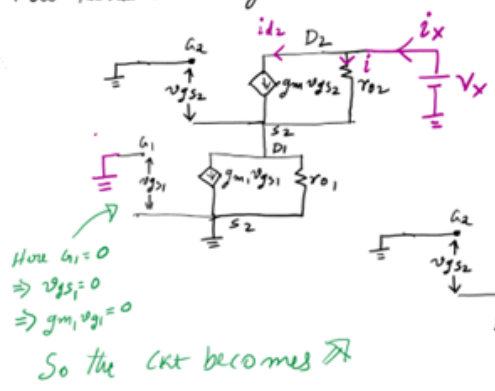
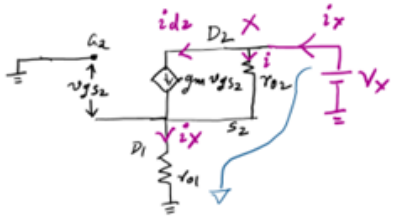
Q4	A first-order active low-pass filter shown in the figure is to be designed with a DC gain of 40 dB and a 3-dB cutoff frequency of 500 Hz. The input resistance should be at least 10 kΩ. Determine a suitable set of component values for R_1, R_2, and C.  1. **R_1 = 10 kΩ, R_2 = 1 MΩ, C = 318 pF 2. R_1 = 10 kΩ, R_2 = 100 kΩ, C = 3.18 nF 3. R_1 = 100 kΩ, R_2 = 1 MΩ, C = 31.8 nF 4. R_1 = 1 kΩ, R_2 = 100 kΩ, C = 3.18 μF	**0.5**	**4**
Q5	An active filter is described by the transfer function H(s) = $-R_2Cs / (R_1Cs + 1)$. Identify the filter type and the slope of its magnitude Bode plot. 1. Low-Pass Filter, -20 dB/decade 2. **High-Pass Filter, +20 dB/decade 3. Band-Pass Filter, -20 dB/decade 4. High-Pass Filter, -20 dB/decade	**0.5**	**4**
Q6	An amplifier has R_i = 100 kΩ, Aoc = 100 V/V, and R_0 = 1 Ω. It is driven by a source with R_S = 25 kΩ and drives a load R_L = 3 Ω. Calculate the overall voltage gain V_0/V_S 1. 45 V/V 2. **60 V/V 3. 75 V/V 4. 100 V/V	**0.5**	**4**
Q7	An ideal op-amp differentiator shown in the figure has a feedback resistor R = 20 kΩ and an input capacitor C = 10 nF. The circuit is supplied with a symmetric triangular wave input that has a peak-to-peak voltage of 4 V and a period of 2 ms. What is the output waveform and its peak voltage 1. A sinusoidal wave with a peak voltage of ±1.26 V 2. A triangular wave with a peak voltage of ±0.4 V 3. A square wave with a voltage level of ±0.4 V 4. **A square wave with a voltage level of ±0.8 V	**0.5**	**4**

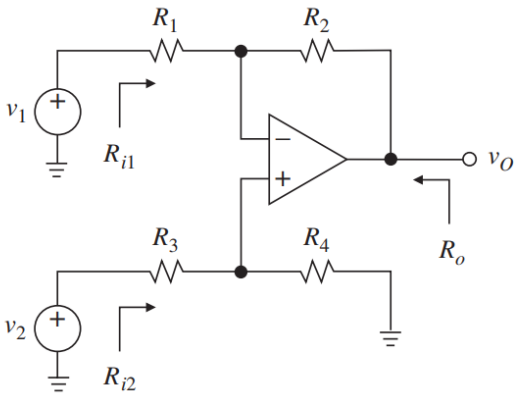
Q8	For a circuit to be stable, where must all poles and zeros be located in the s-plane 1. **Poles in the left half-plane, Zeros can be anywhere. 2. Poles in the right half-plane, Zeros in the left half-plane. 3. Both poles and zeros in the right half-plane. 4. Poles can be anywhere, only zeros must be in the left half-plane	**0.5**	**4**
Q9	A MOSFET is biased at a drain current of 0.5 mA. If μnCox = 100 μA/V2, W/L = 10, and λ = 0.1 V-1, intrinsic gain of the device is_____ 1. 10 2. **20 3. 5 4. 15	**0.5**	**1**
Q10	The voltage gain for the circuit shown below is _____. Consider λ≠0. 1. -(gm1×gm2)(ro1||ro2) 2. **-(gm1+gm2)(ro1||ro2) 3. -(gm1)(ro1||ro2) 4. -(gm2)(ro1||ro2)	**0.5**	**3**
Q11	Design the circuit shown in the figure by obtaining R1, R2, R3 and R4 for I_D = 0.5 mA, V_{th} = 1 V, $\mu_n C_{OX}$ W/L = 1 mA/V^2, a power budget of 7.5mW, and an input impedance of 250 kΩ . Assume $V_D = (2V_{DD}/3)$, VS = $(V_{DD}/3)$ and λ = 0. Draw the small signal model of the circuit  	**(4M)**	**3**

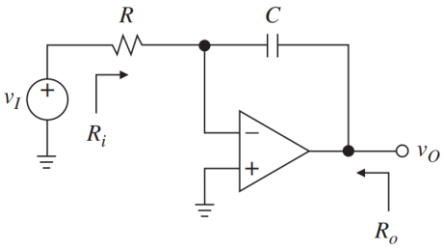
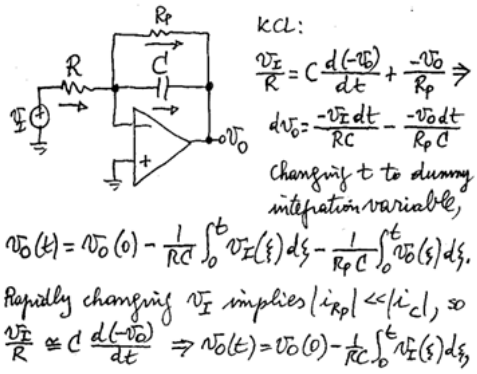
			
Q12	Illustrate the following MOS biasing techniques a) fixed bias with source resistance b) drain feedback c) constant current (with current mirror). Discuss the advantages and disadvantages associated with each of these biasing techniques.	**(3M)**	**3**

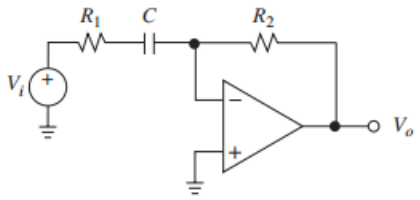
	(a) Fixing V_G with source resistance  • We can write $V_G = V_{GS} + R_S I_D$ • Here Rs is negative feedback resistor. • Assume by some reason I_D increases. • From the equation we know that V_G is constant. • So V_{GS} must decrease. ➔This intern results in decrease I_D. • So Rs is negative feedback provider which is also known as degeneration resistance.	**0.5M**	
	Advantages: • Better stability compared to simple fixed bias (without RSR_SRS). • Easy to design. Disadvantages: • Voltage drop across Rs reduces voltage headroom at drain. • Power dissipation in RS reduces efficiency	**0.5M**	
	b) Drain feedback  • R_G is large register (MΩ) $V_{GS} = V_{DS} = V_{DD} - R_D I_D$ $V_{DD} = V_{GS} + R_D I_D$ • If I_D increases, V_{DD} being constant, V_{GS} must reduce. • ➔ So if V_{GS} reduce I_D reduces • R_G is negative feedback resistor	**0.5M**	
	Advantages **Improves bias stability**: • Compensates for variations in Vth, k, or temperature. • Limits large changes in drain current. **Provides negative feedback**: • Makes circuit more predictable.	**0.5M**	

	Disadvantages • **Consumes voltage headroom**: Drop across RSR_SRS reduces VDS, limiting output swing. • **Extra power dissipation**:		
	C) Constant current (with current mirror). **Biasing by constant current source (with current Mirror)** • The most effective scheme for biasing • R_G is large register established DC GND • R_D establishes DC voltage to allow for the required output signal swing while ensuring that the transistor always remains in the saturation region. 	**0.5M**	
	Advantages • **Stable and Predictable Bias Current** Ensures excellent **bias point stability** compared to resistor biasing. • **High Gain Possibility** Provides better performance for analog circuits like differential pairs, op-amps. **Dis-Advantages** • **Higher Complexity & Area** • Requires extra transistors (reference and mirror devices). • Consumes more silicon area compared to simple resistor biasing. • **Limited by Mismatch & Channel-Length Modulation** • Current mirror accuracy depends on transistor matching. • Channel-length modulation (λ\lambdaλ) causes finite output resistance → bias current is not perfectly constant.	**0.5M**	
Q13	Explain the importance of the circuit shown in the figure. Using Small signal model derive R_{in} and R_{out} by assuming λ≠0 	**(3M)**	**3**

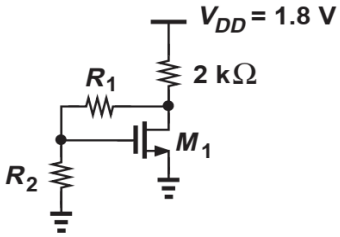
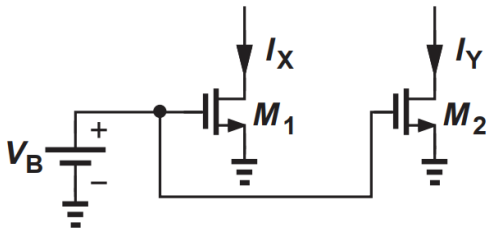
	 With CG Amp we get ① High I/P Impedance ② Large transconductance ③ Superior high freq/. response 	0.5M	
	 At the I/P Section, $R_{in} = \infty$	1.5M	
	Now redraw the small signal model,  Here $G_1 = 0$ $\Rightarrow v_{gs_1} = 0$ $\Rightarrow g_{m_1} v_{g_1} = 0$  So the ckt becomes	1.5M	

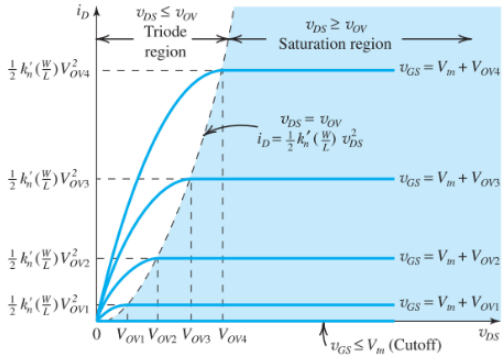
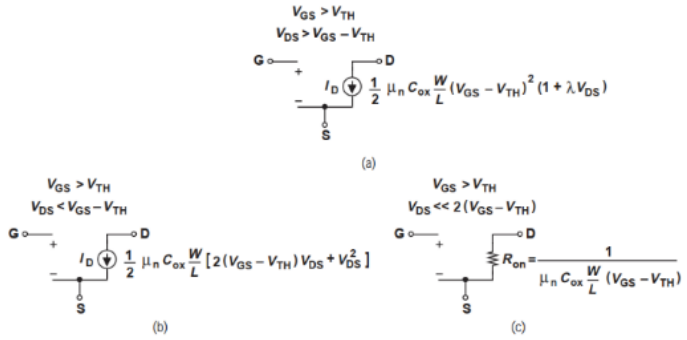
	At node x, $i_x = i + i_{d_2}$ ($i_{d_2} = g_{m_2} v_{gs_2}$) $i = i_x - i_{d_2}$ ——— ① Apply KVL @ o/p Section $V_x = i r_{o_2} + i_x r_{o_1}$ sub for i from ① $V_x = [i_x - g_m v_{gs_2}] r_{o_2} + i_x r_{o_1}$ — ② See now, v_{gs_2} is appearing als r_{o_1} $\Rightarrow v_{gs_2} = -v_{ds_1} = -i_x r_{o_1}$ Now $v_{gs_2} = -i_x r_{o_1}$ in ② $V_x = i_x r_{o_2} + i_x g_{m_2} r_{o_1} r_{o_2} + i_x r_{o_1}$ $\frac{V_x}{i_x} = r_{o_2} + r_{o_1} + g_{m_2} r_{o_1} r_{o_2}$ Larger than rest of the terms. $\therefore R_{out} = \frac{V_x}{i_x} = g_{m_2} r_{o_1} r_{o_2}$		
Q14	For the circuit shown in the figure, R1 = R3 = R4 = 10 k Ω and R2 = 30 k Ω. (a) If V_1 = 3 V, find V_2 for V_0 = 10 V. (b) If V_2 = 6 V, find V_1 for V_0 = 0 V. (c) If V_1 = 1 V, find the range of values for V_2 for which −10 V ≤ V_0 ≤ +10 V  	**(3M)**	**4**

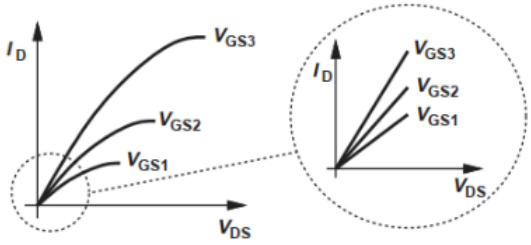
	$v_O = \frac{R_2}{R_1}\left(\frac{1 + R_1/R_2}{1 + R_3/R_4}v_2 - v_1\right)$ Substituting values and solving Vo= 2v2 -3v1 Part (a): $v_1 = 3$ V, find v_2 for $v_o = 10$ V Using the transfer function: $10 = 2v_2 - 3(3)$ $10 = 2v_2 - 9$ $v_2 = 9.5$ V	1M	
	Part (b): $v_2 = 6$ V, find v_1 for $v_o = 0$ V $0 = 2(6) - 3v_1$ $0 = 12 - 3v_1$ $v_1 = 4$ V	1M	
	Part (c): $v_1 = 1$ V, find v_2 range for $-10\text{ V} \le v_o \le +10\text{ V}$ With $v_1 = 1$ V: $v_o = 2v_2 - 3$ For $v_o = -10$ V: $v_2 = -3.5$ V For $v_o = +10$ V: $v_2 = 6.5$ V Answer: $-3.5\text{ V} \le v_2 \le 6.5$ V	1M	
Q15	Derive the relationship between V_0 and V_i if the circuit shown in the figure includes a resistance R_P in parallel with C. Discuss the extreme cases of V_i changing very rapidly and very slowly 	**(3M)**	**4**
		Circuit & Vo(t) derivation=1M Slow & rapidly changing Vi discussion= 1M each	

	indicating integrator behavior. Slowly changing v_I implies $\lvert i_C \rvert \ll \lvert i_{R_p} \rvert$, so $\frac{v_I}{R} \cong -\frac{v_O}{R_p}$, or $v_O \cong \left(-\frac{R_p}{R}\right) v_I$, indicating amplifier behavior.		
Q16	In the circuit of figure, specify suitable components to achieve a -3dB frequency of 500 Hz with a DC gain of 26 dB and an input resistance of at least 5 kΩ. (b) At what frequency does gain drop to 6 dB? What is the phase there  	**(3M)**	**4**
	$G_{dB} = 20\log_{10}\left(\frac{V_{out}}{V_{in}}\right)$ $\frac{V_{out}}{V_{in}} = 10^{G_{dB}/20}$. **Relevant formulas (inverting first-order high-pass)** For the inverting high-pass (input: series C then R_1; feedback: R_2) the transfer fun $$H(s) = -\frac{R_2}{R_1}\frac{sR_1C}{1+sR_1C} = -\frac{R_2}{R_1}\frac{s/\omega_c}{1+s/\omega_c},$$ where $$\omega_c = \frac{1}{R_1C}, \qquad f_c = \frac{\omega_c}{2\pi} = \frac{1}{2\pi R_1C}.$$ The high-frequency (passband) linear gain is $$G_\infty = \frac{R_2}{R_1}.$$ Magnitude (using $x = \frac{f}{f_c}$): **(a) Choose components** **Requirement 1 — gain:** 26 dB → linear gain $$G_\infty = 10^{26/20} = 10^{1.3} \approx 19.95 \approx 19.95.$$ So $R_2/R_1 \approx 19.95$. **Requirement 2 — input resistance:** For high frequencies the input impedance ≈ R_1. We need $R_1 \geq 5\ \text{k}\Omega$. Choose the minimum practical value to keep R_2 reasonable. Let $$\boxed{R_1 = 5\ \text{k}\Omega.}$$ Then $$R_2 = 19.95 \times 5\ \text{k}\Omega \approx 99.76\ \text{k}\Omega \approx 100\ \text{k}\Omega\ (\text{standard}).$$ **Requirement 3 — corner $f_c = 500$ Hz:** $$C = \frac{1}{2\pi f_c R_1} = \frac{1}{2\pi \cdot 500 \cdot 5000} = \frac{1}{2\pi \cdot 2.5 \times 10^6} \approx 6.366 \times 10^{-8}\ \text{F}.$$ So $$C \approx 63.66\ \text{nF}.$$ Choose a standard capacitor near this value. Common standard values: 62 nF or 68 nF. Choose the closer common value, e.g. $$\boxed{C = 62\ \text{nF}}.$$ **Resulting practical component set** $$R_1 = 5.0\ \text{k}\Omega, \quad R_2 = 100\ \text{k}\Omega, \quad C = 62\ \text{nF}.$$	**1M**	

	Resulting practical component set $$R_1 = 5.0\,\text{k}\Omega, \quad R_2 = 100\,\text{k}\Omega, \quad C = 62\,\text{nF}.$$ **Check actual cutoff with chosen C:** $$f_c = \frac{1}{2\pi R_1 C} = \frac{1}{2\pi \cdot 5 \times 10^3 \cdot 62 \times 10^{-9}} \approx 513.3\,\text{Hz},$$ which is acceptably close to 500 Hz. **(b) Frequency where gain = 6 dB and the phase there** **Step 1 — convert 6 dB to linear scale** $$G_{6\text{dB}} = 10^{6/20} \approx 10^{0.3} \approx 1.995 \approx 1.995.$$ **Step 2 — set magnitude equation** Let $H_0 = R_2/R_1 \approx 19.95$. Using $x = f/f_c$, $$H_0 \frac{x}{\sqrt{1+x^2}} = 1.995.$$ Solve for x. Square both sides: $$H_0^2 \frac{x^2}{1+x^2} = (1.995)^2 \approx 3.98.$$ $$(H_0^2 - 3.98)\,x^2 = 3.98 \quad \Rightarrow \quad x^2 = \frac{3.98}{H_0^2 - 3.98}.$$ Using $H_0 \approx 19.95 \Rightarrow H_0^2 \approx 398.0$, $$x^2 = \frac{3.98}{398.0 - 3.98} \approx \frac{3.98}{394.02} \approx 0.01010,$$ $$x \approx \sqrt{0.01010} \approx 0.1005.$$ So the frequency is $$f_{6\text{dB}} = x\, f_c \approx 0.1005 \times f_c.$$ Using the target $f_c = 500$ Hz: $$f_{6\text{dB}} \approx 0.1005 \times 500 \approx 50.3\,\text{Hz}.$$ Using the practical value $f_c \approx 513.3$ Hz (with C=62nF): $$f_{6\text{dB}} \approx 0.1005 \times 513.3 \approx 51.6\,\text{Hz}.$$ Either is fine; the design target answer is about **50 Hz** (≈51.6 Hz with chosen standard C). **Step 3 — phase at that frequency**	**1M**	
	Phase formula: $$\angle H(\omega) = -90^\circ - \tan^{-1}(x).$$ With $x \approx 0.1005$, $$\tan^{-1}(0.1005) \approx 5.76^\circ,$$ so $$\angle H \approx -90^\circ - 5.76^\circ = -95.76^\circ.$$ (Using f_c = 513.3 Hz gives essentially same angle since x same: $\approx -95.8^\circ$.)	**1M**	

Q17	In the circuit shown, below the drain current (IDS1) in MOSFET (M1) is 0.5 mA. If W/L = 50/0.18, μnCox = 100 μA/V2, VTH = 0.5 V, λ = 0. Calculate the values of R1 and R2 such that these resistors carry a current equal to one-tenth of IDS1. Consider that the M1 is operating in saturation 	**(2M)**	**3**
	$I_{R1,R2}=I_{DS1}/10 =0.05mA$ $I_{DS1}=0.5\ \mu_n C_{ox}(W/L)(V_{GS1}-V_{TH})^2$ $0.5\times10^{-3}=0.5\times100\times10^{-6}(50/0.18)\ (V_{GS1}-0.5)^2$ $V_{GS1}=0.689V$ $R_2= 0.689/ 0.05\times10^{-3}$ **$R_2=13.78K\Omega$**	**1M**	
	$1.8- 0.55\times10^{-3}\times2\times10^{3} =V_{D1}$ $V_{D1}=0.7V$ $I_{R1}= V_{D1}- V_{GS1}/R_1$ $R_1= V_{D1}- V_{GS1}/ I_{R1}$ $R_1=0.7-0.689/0.05\times10^{-3}$ **$R_1=220\Omega$**	**`1M**	
Q18	For the circuit shown in the figure, M1 and M2 have lengths equal to 0.25 μm and λ = 0.1 V^{-1}. Determine W1 and W2 such that $I_X = 2I_Y = 1$ mA. Assume $V_{DS1} = V_{DS2} = V_B = 0.8$ V. What is the output resistance of each current source 	**2M**	**3**
	For M_1, $I_x = \frac{1}{2}(200\times10^{-6})(\frac{W_1}{0.25})(0.8-0.4)^2(1+0.1(0.8))$ $10^{-3} = 0.16\times10^{-4}(\frac{W_1}{0.25})(1.08)$ $\therefore W_1 \approx 14.5\mu$ // For M_2, $0.5\times10^{-3} = 0.16\times10^{-4}(\frac{W_2}{0.25})(1.08)$ $\therefore W_2 = 7.25\mu$ //	**1M**	

	Output resistance $= r_o$ $= \frac{1}{\lambda} \times \frac{1}{I_D}$ $\therefore \quad r_{o1} = \left(\frac{1}{0.1}\right)\left(\frac{1}{10^{-3}}\right)$ $= 10\,k\Omega //$ $r_{o2} = \left(\frac{1}{0.1}\right)\left(\frac{1}{0.5\times10^{-3}}\right)$ $= 20\,k\Omega //$	**1M**	
Q19	Draw the output characteristics of NMOS and show the saturation region, triode and deep triode regions. Show the large signal model in all those regions	**(2M)**	**1**
	  Figure 6.29 MOS models for (a) saturation region, (b) triode region, (c) deep triode region.	**Diagram = 1M** **Model & Eq. = 1M**	

	 I_D V_{GS3} V_{GS2} V_{GS1} V_{DS} I_D V_{GS3} V_{GS2} V_{GS1} V_{DS}		